Rygel Yance

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Education:

B.S. in Computer Science | B.S. in Mathematics | University of Maryland, College Park *May 2025*

Experience:

Immersive Research Internship Experience (IRIE) | GitHub Page

Undergraduate Research Assistant *Summer 2024*

- **What:** Worked with Assistant Clinical Professor Thanicha Ruangmas to analyze how the opening of the METRO Green & Blue light rail lines affected demographic groups in Minneapolis & St. Paul, MN.
- **Used:** R, GitHub, tidyverse, tidycensus, tigris, terra, dplyr, ggplot2, maptiles, knitr
- **Data Collection:** Queried PM2.5 data from NASA Earth, meteorological data from the GLDAS Catchment Land Surface Model L4.
- **Results:** Found that the opening of the light rail led to between a 15-18% reduction in PM2.5 air pollution across all age groups, with younger age groups seeing slightly higher reductions.

Southern Economic Association Conference Presentation | Program Info.

Main Presenter of "Do Light Rails Provide a Track to Cleaner Air?" *Fall 2024*

- **What:** Worked with Assistant Clinical Professor Thanicha Ruangmas to finalize and present our research on the effect of light rail openings on air pollution at the 2024 annual SEA meeting. Served as the main presenter with Landon Thomas, another undergraduate research assistant.
- **Used:** Difference-in-Differences statistical model, LaTeX Slide formatting, R.
- **Results:** Found that the light rail cities would reduce air pollution on high traffic days when analyzing individual cities, but no general trend existed when looking across all cities. Gained valuable insight into future avenues of research and investigation on this topic.

Relevant Coursework:

Student, University of Maryland

Computer Science – Machine Learning Track *2022-2025*

- Introduction to Machine Learning
- Introduction to Data Science

Mathematics – Statistics Track *2022-2025*

- Applied Statistics & Probability I & II
- Calculus I, II, II & Advanced Calculus I

Programming Skills:

- **Languages (Sorted by proficiency):**
  - **Advanced:** Java, C, R, Python
  - **Intermediate:** OCaml, Rust, Matlab, SAS
  - **Beginner:** MIPS Assembly, HTML, CSS, JavaScript
- **Technologies:** Git, SQLite, LaTeX